

Question ID: 5ae186b4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$\frac{-54}{w} = 6$$

What is the solution to the given equation?

Correct Answer: -9

Rationale

The correct answer is **−9**. Since *w* is in the denominator of a fraction in the given equation, *w* can't be equal to **0**. Since *w* isn't equal to **0**, multiplying both sides of the given equation by *w* yields an equivalent equation, **−54 = 6*w***. Dividing both sides of this equation by **6** yields **−9 = *w***. Therefore, **−9** is the solution to the given equation.